SciLifeLab

essence
THE e-SCIENCE COLLABORATION

AI Agents in Life Sciences

Practical Tutorials into AI Agents

shorturl.at/rmCjV

Scan to get the workshop code link

AI Agent in Life Science with LangGraph

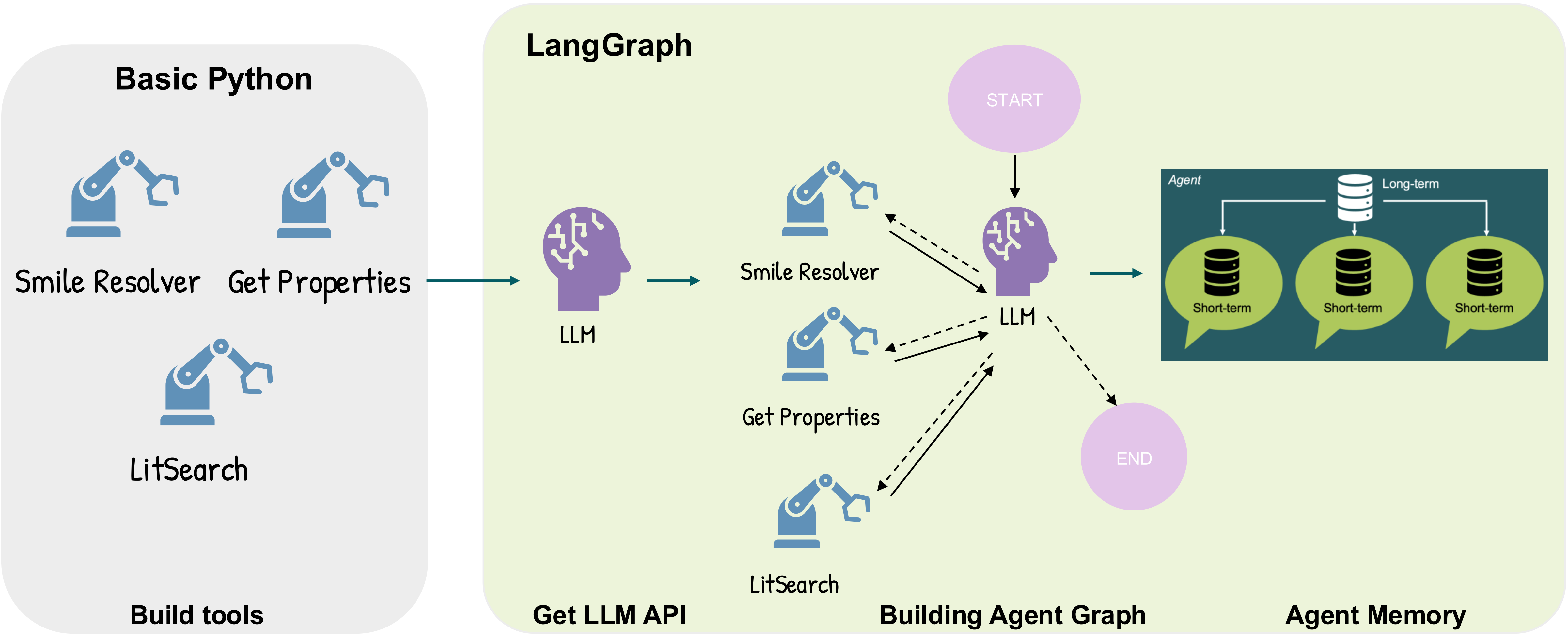
”Hello World” – An introduction and tutorial

Presenter:

- *Dinh Long Huynh, PhD Student at Uppsala University*

Scan to get the workshop code link

What are we going to do today?



Nucleic Acids Research **53**, W1, 2025 *Nucleic Acids Research* ***43**, W1, 2015*

Workshop outline

- **Part 1:** Setup & imports
- **Part 2:** Understanding and creating tools

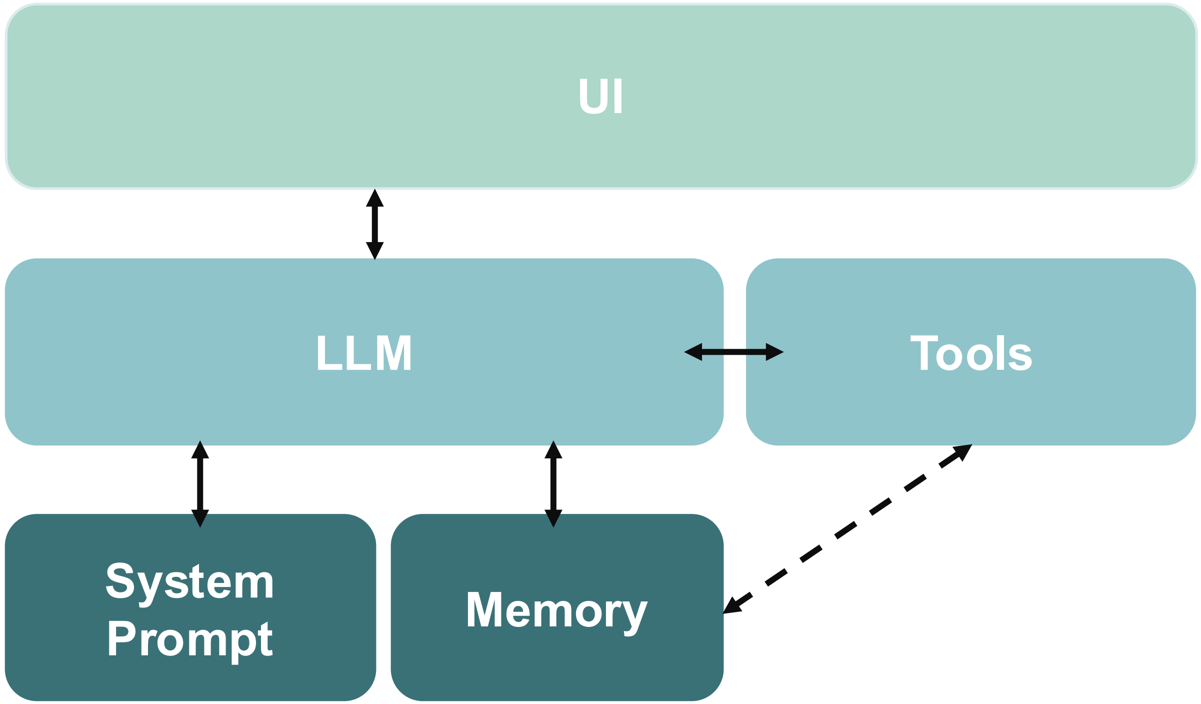
- **Part 3:** Defining the state
- **Part 4:** Building the agent graph
- **Part 5:** Testing the agent

- **Part 6:** Adding short-term memory

- **Part 7:** Exploring prebuilt agents

- **Part 8 (optional):** Extension exercises on CodeAct and multi-agent system

What is Agentic AI system?



1. Autonomous software system
2. Perceives environment and context
3. Makes decisions
4. Takes actions
5. Achieve specific goals
6. Minimal human intervention

About LangChain ecosystem

LangChain

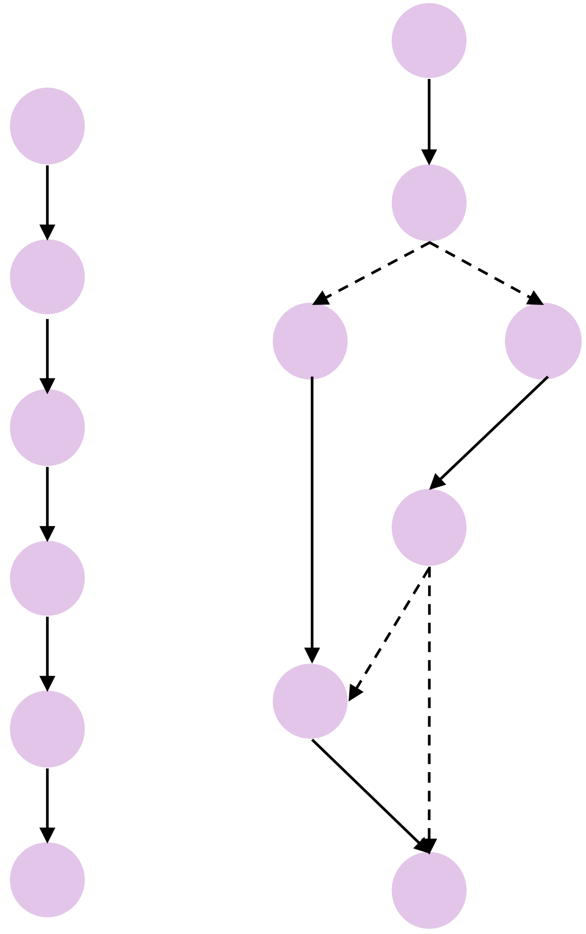
Build chain workflow

LangGraph

Build graph workflow

LangSmith

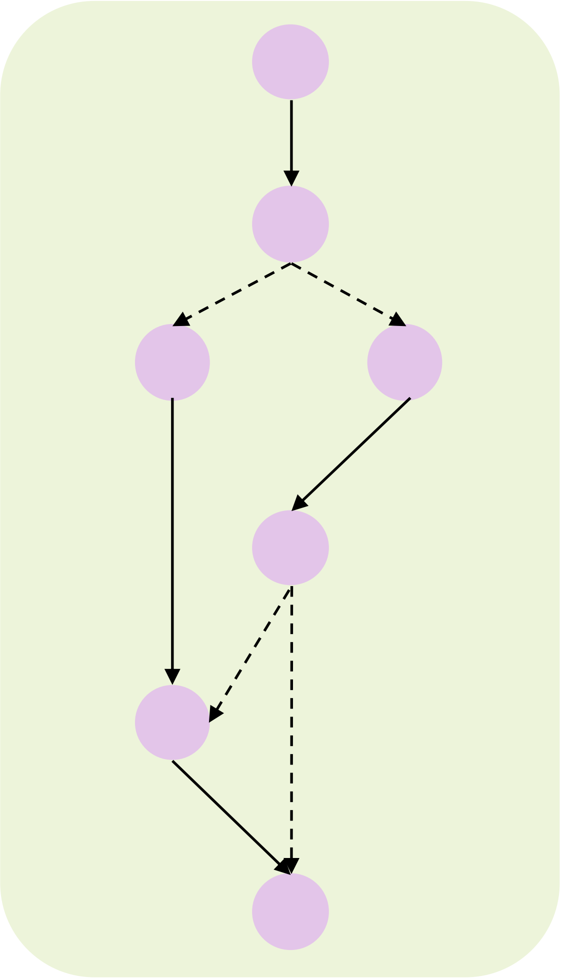
Trace running process of all LangChain’s product



Chain Graph

LangChain Documentation, 2025

Basic Concepts in LangGraph



Graph

Node

A unit of computation: function, tool, pipeline or agent

Edge

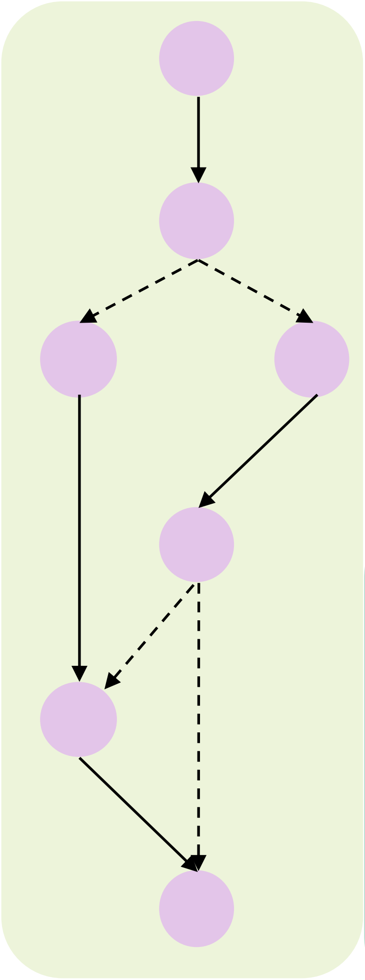
Static edge — A single compulsory route

Conditional edges — A flexible route, usually require **decision making** from logical function (if-else), LLM or human

State

Shared/Persistent context across the graph.
After node invocation, it updates the States.

State in LangGraph



Graph

State Shared/Persistent context across all nodes. After node invocation, it updates the States.

```
from typing_extensions import TypedDict
from typing import Annotated, List, Dict
from langgraph.graph.message import add_messages


class ChatState(TypedDict):
        messages: Annotated[List[Dict], add_messages]
```

State

messages = [
{‘human_message’: Hi, How are you today?}
]

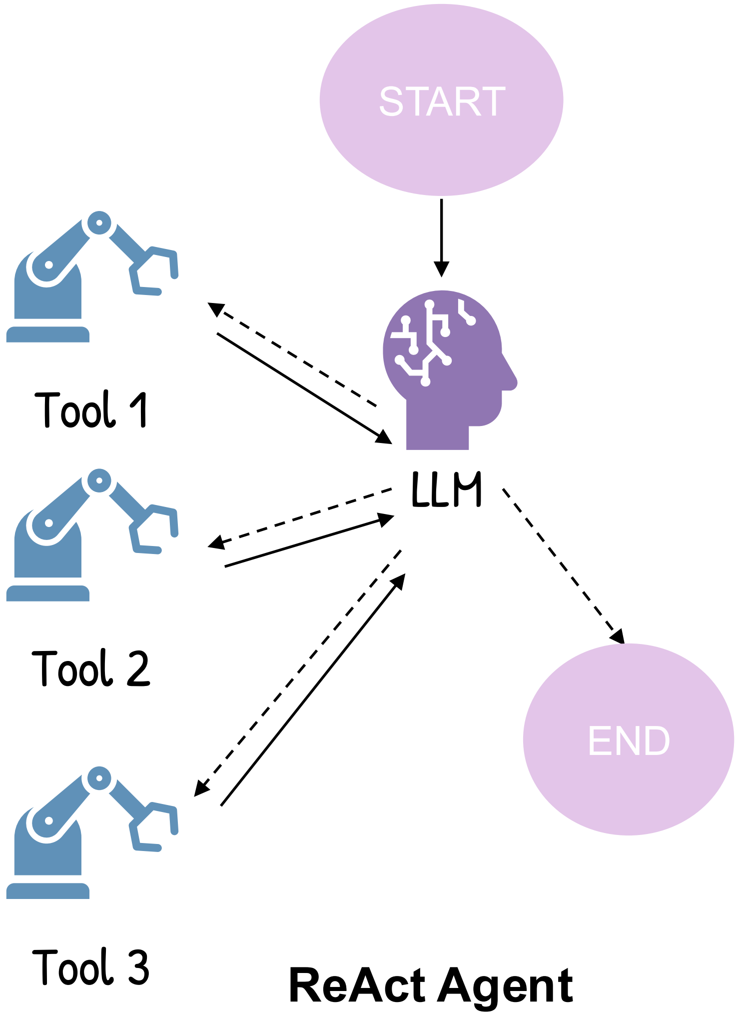
State

messages = [
{‘human_message’: Hi, How are you today?},
{‘ai_message’: I am good, what you need me to do?}
]

State

messages = [
{‘human_message’: ‘Hi, How are you today?’},
{‘ai_message’: ‘I am good, what you need me to do?’},
{‘human_message’: ‘Please find me 10 antiviral drugs that can treat COVID-19.’}
]

AI Agent - A simple but flexible graph

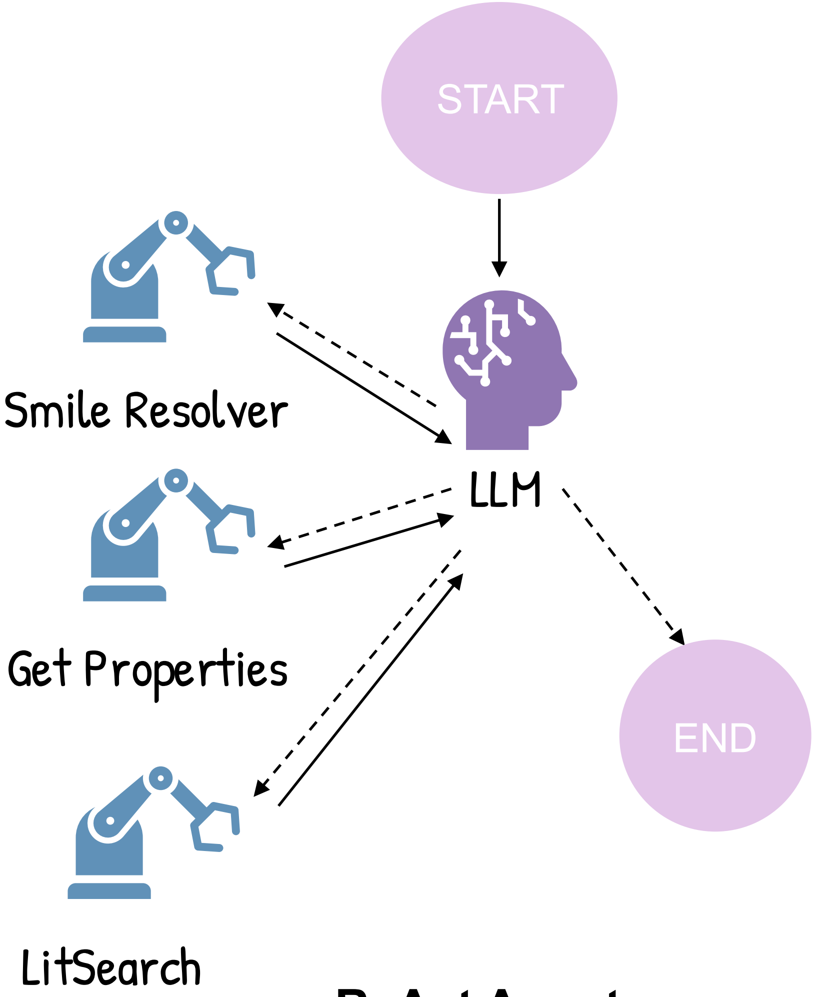


ReAct Agent

ReAct Agent: Reasoning – Acting Agent

- Flexible graph with many conditional edges with a LLM in the center.
- LLM determines which tools to use
- LLM determines when to continue or terminate

AI Agent – Demo:



ReAct Agent

SMILES Resolver: resolve any compounds identifers into SMILES string.

Get Properties: use RDKit to get basic physiochemical properties of compounds, i.e. MW, logP, HBA, HBD, etc.

LitSearch: get relevant passages among all PubMed publication given user's query

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Prompts:

” You are an expert drug discovery researcher. Use your available tools to answer the user’s question as accurately as possible. Never fabricate or invent data. Here is the compound ID: Erlotinib. Is it orally drug-like?”

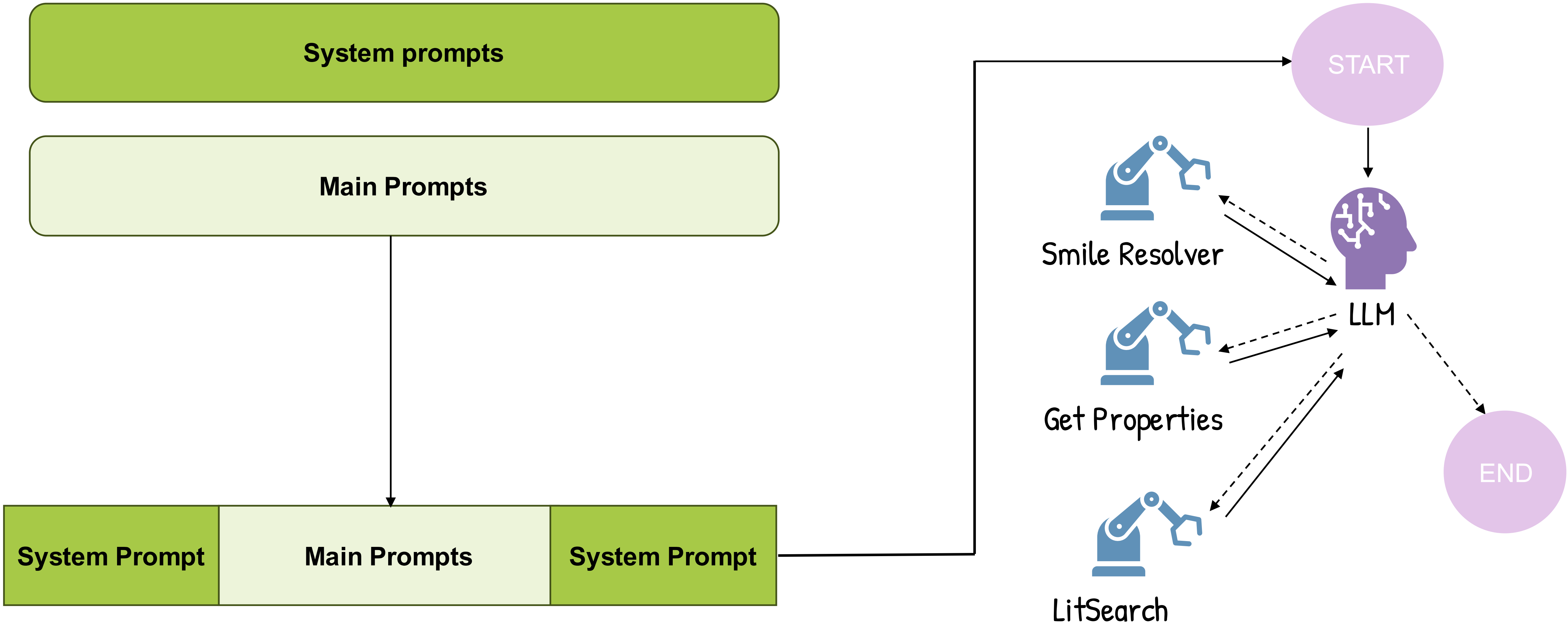
" You are an expert drug discovery researcher. Use your available tools to answer the user’s question as accurately as possible. Never fabricate or invent data. Which is more lipophilic: Imatinib or Dasatinib?”

" You are an expert drug discovery researcher. Use your available tools to answer the user’s question as accurately as possible. Never fabricate or invent data. Check betulinic acid. Are there any structural red flags for oral bioavailability?”

" You are an expert drug discovery researcher. Use your available tools to answer the user’s question as accurately as possible. Never fabricate or invent data. What are known EGFR inhibitors in the literature, and are they drug-like?”

" You are an expert drug discovery researcher. Use your available tools to answer the user’s question as accurately as possible. Never fabricate or invent data. What are the common physicochemical characteristics of successful JAK inhibitors?"

System Prompts:

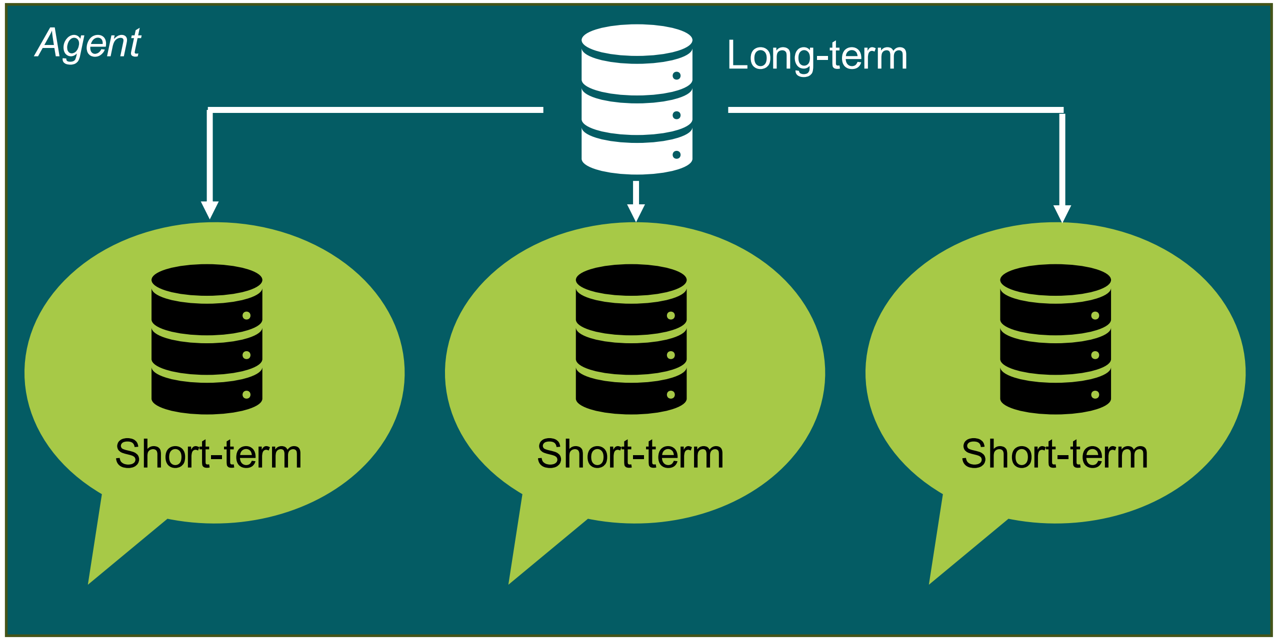


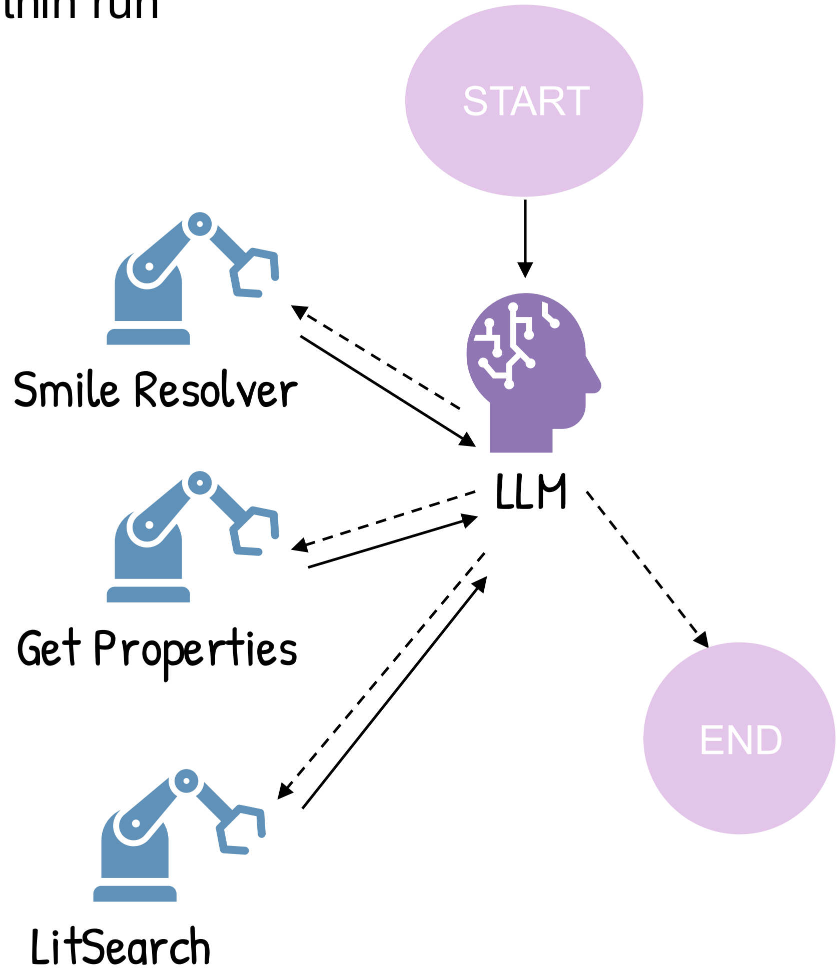
Memory:

- GraphState only maintains context from START to END: Memory within run

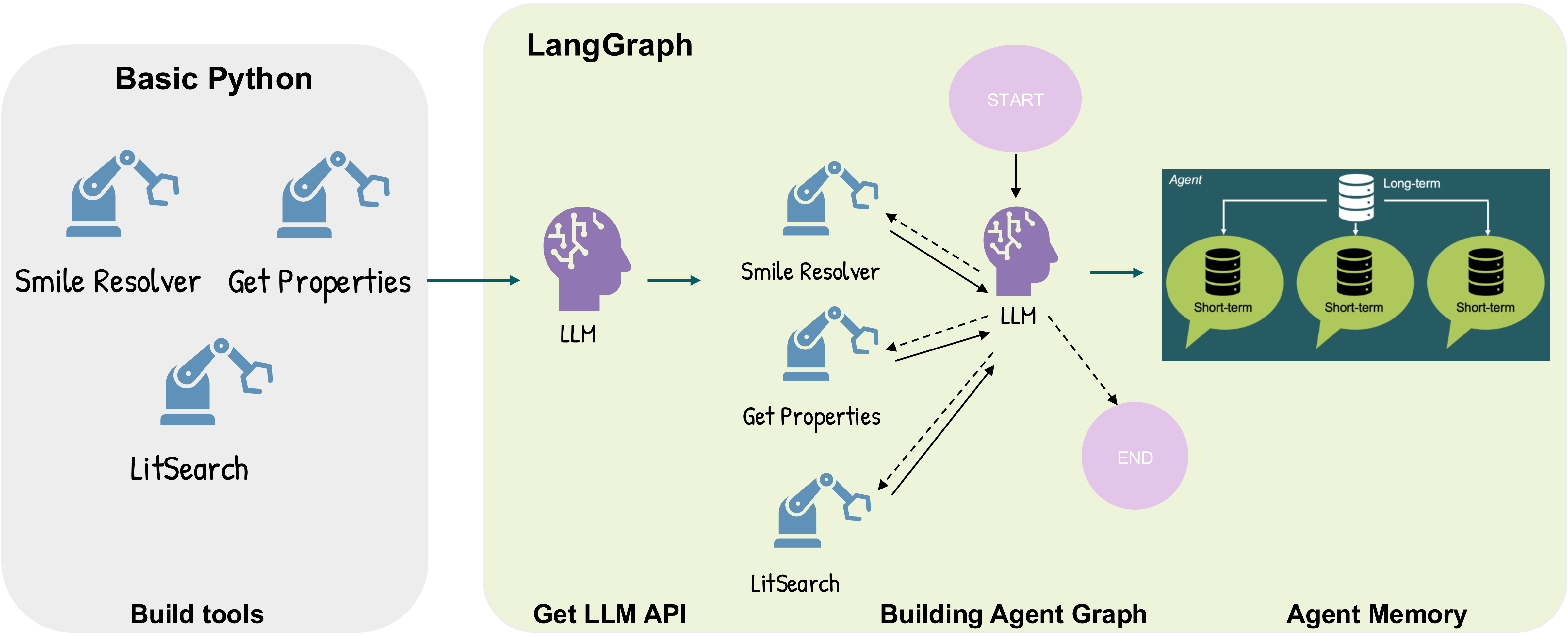
- Short-term memory: Memory within conversation.

- Long-term memory: Memory outside conversation.





What are we going to do today?



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Problem with ReAct & Tools

Pre-defined tool is rigid

CCC(CC)COC(=O)C(C)NP(=O)(OCC1C(C(C(O1)(C#N)C2=CC=C3N2N=CN=C3N)O)O)OC4=CC=CC=C4

Remdesivir

CID 121304016

```
@tool
def awesome_tool(smiles: str):
…
```

Fixed tool space

Tool 1

Tool 4

Tool 2

Tool 3

Scaling: Agent works well with 3-4 tools does not means it works well with 30-40 tools

```
@tool
def tool1():
    ''' Tool description '''
```

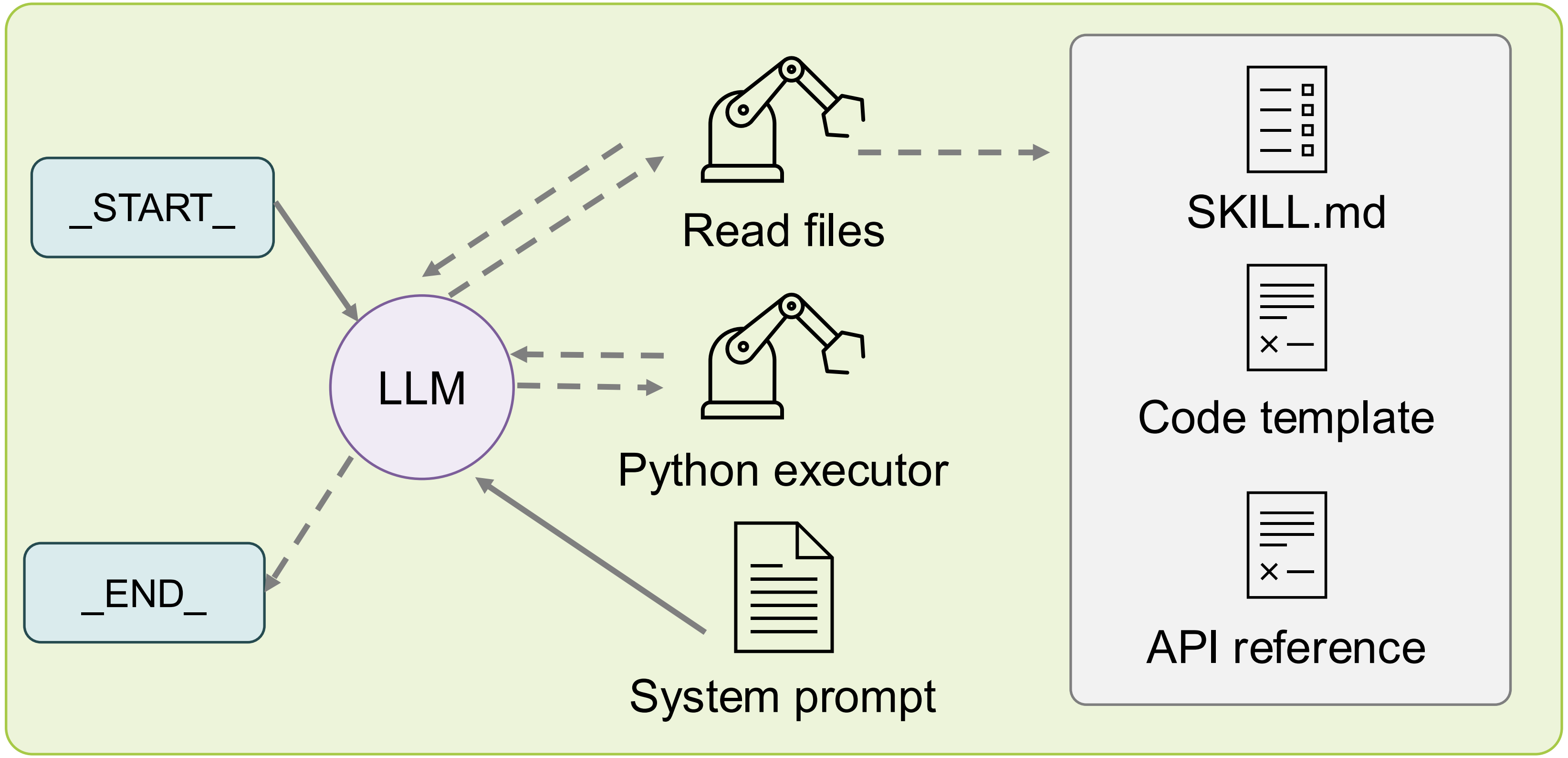
```
@tool
def tool2():
    ''' Tool description '''
```

llm.bind_tool(TOOLS)

System prompt

Available tools:
tool1: Description
tool2: Description

CodeAct Agent

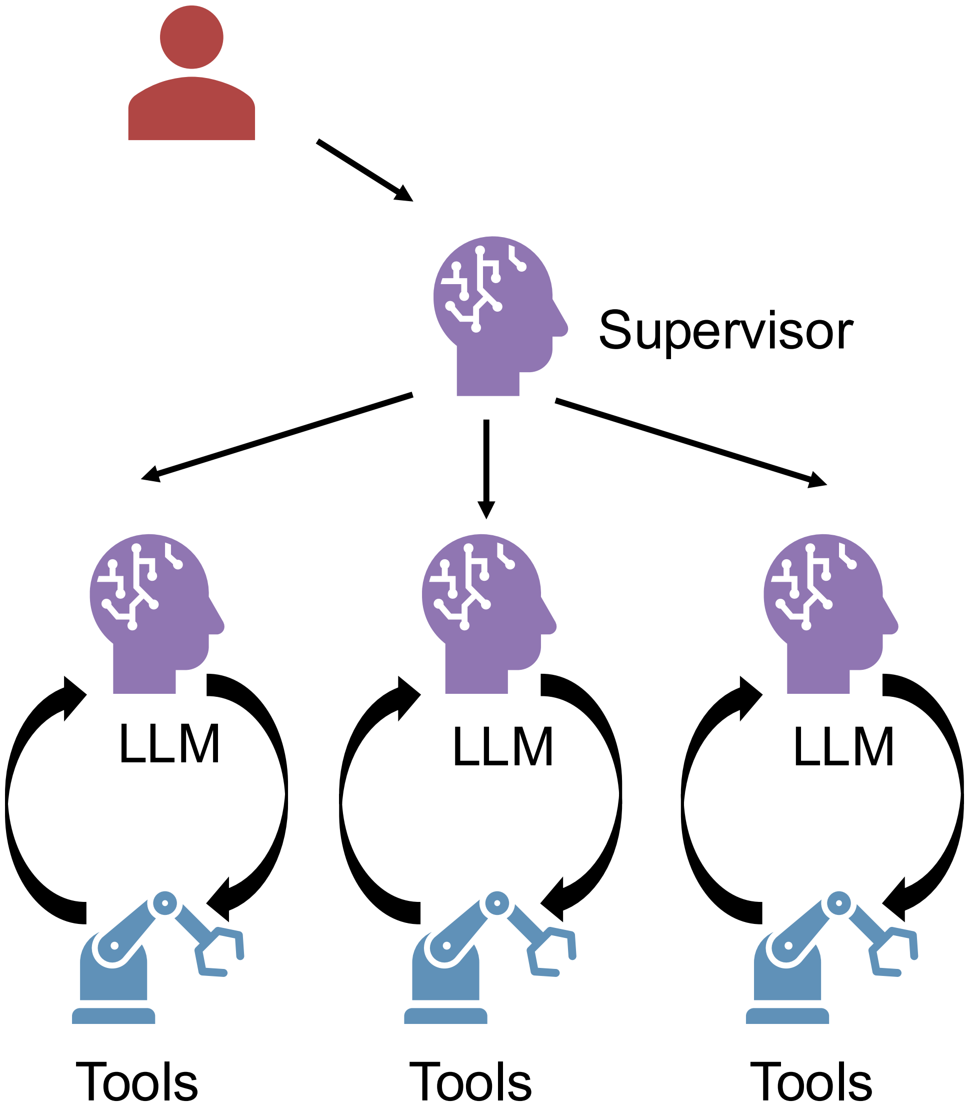


Conditional step, determined by the LLM's reasoning

Mandatory step

Wang, X. et. al, *Executable code actions elicit better LLM agents* (arXiv:2402.01030). arXiv. https://arxiv.org/abs/2402.01030

Multi-agentic AI systems



Supervisor system

References:

1. Docs by LangChain, 2025, from https://docs.langchain.com
2. Yeganova, L., Kim, W., Tian, S., Comeau, D. C., Wilbur, W. J., & Lu, Z. (2025). LitSense 2.0: AI-powered biomedical information retrieval with sentence and passage level knowledge discovery. Nucleic Acids Research, 53(W1), W361–W368
3. Overington, J. P., et al. (2015). ChEMBL web services: streamlining access to drug discovery data and workflows. Nucleic Acids Research, 43(W1), W612–W620.

Thank you for listening!

Feedbacks and Question